

GSE Risk Sharing Deal Review

Primer: GSE Risk Sharing Transactions

- We have discussed GSE risk sharing transactions several times over the course of 2013. This primer compiles these discussions into one stand-alone publication, and covers many aspects of these transactions including issuance, structure, collateral, relative value, and other applications of pricing data. Now that the proof of concept of transferring credit risk to the private market has been established, we expect to see more programmatic issuance over the course of 2014.
- **Highlights:**
 - While overall issuance seems low, the GSE transactions represent a significant share of mortgage credit risk sold over 2013.
 - The deals are debt securities, bonds sold are uncapped LIBOR floaters, and cash flows are based on a reference pool of assets.
 - Money managers and insurance companies are natural buyers, while banks may not find this an attractive investment given the thinness of the tranches.
 - The collateral is substantially similar, consisting of post-crisis origination, 30-year, amortizing fixed-rate loans with LTVs in the 70-80% area.
 - Trading price action as well as deal pricing has been stable, in part due to transparent trade information.

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Primer—GSE Risk Sharing Transactions

Background

The GSEs first transferred credit risk in 1998.

The notion of transferring credit risk from Fannie/Freddie to the private market is not new. Freddie Mac priced a risk sharing transaction in May 1998 (called Mortgage Default Recourse Notes, or MODERNS), in which they sold subordinate credit classes whose payments and losses were assigned based on a reference pool of assets. Loans were “bought out” at a certain point of delinquency and loans were assigned a fixed loss severity, which were then assigned to bonds. This transaction is similar to the more recent transactions in 2013, and was a likely starting point for creating the new model for transferring credit risk to the private market.

The FHFA's conservatorship scorecard required Fannie and Freddie to transfer credit risk to the private market.

During the post-conservatorship period, the goal of transferring Fannie Mae and Freddie Mac credit risk to the private market was first mentioned on March 9, 2012 as part of the Conservatorship Scorecard published by the Federal Housing Finance Agency (FHFA). In the scorecard, the FHFA directed the GSEs to complete a risk sharing transaction by September 2012. The GSEs were unable to complete a transaction in 2012. Again on March 4, 2013, Acting Director of the FHFA Edward DeMarco published Conservatorship Scorecard performance goals for 2013. Contracting the Government Sponsored Entities (GSEs) market dominance received the highest overall weighting, and the first item of this goal stated the following:

Single Family - Each Enterprise will demonstrate the viability of multiple types of risk transfer transactions involving single family mortgages with at least \$30 billion of unpaid principal balances in 2013. (Risk transfer transactions of less than \$10 billion receive no credit toward this goal, while results between \$10 billion and \$30 billion will receive partial credit.)¹

Other parts of this goal provided metrics for contracting multifamily lending standards as well as shrinking the overall investment portfolio. But this re-iterated, emphasized goal clearly established the requirement for the GSEs to begin selling credit risk to the private market. While the GSEs conducted multiple forms of risk transfer, this article will focus more closely on the securities issued by Fannie (Connecticut Avenue Securities, or CAS, trusts) and Freddie (Structured Agency Credit Risk, STACR, trusts) to sell credit risk.

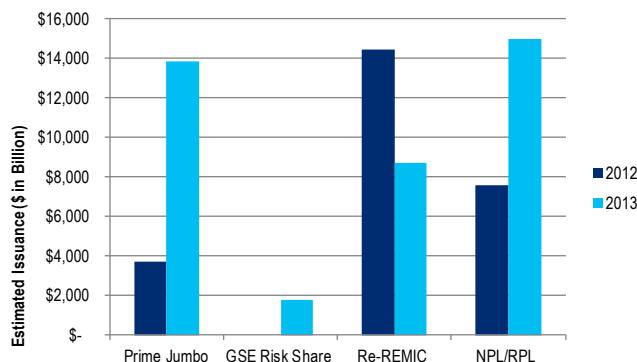
Supply: Issuance Volume in Perspective

2013 issuance was low but represented a sizable portion of credit risk sold.

Figure 1 highlights the relative size of GSE risk sharing transactions to various sub-sectors of non-agency RMBS issuance. During its debut year, the GSEs priced ~\$1.8 billion of securities compared to ~\$13.8 billion of prime jumbo RMBS. While the risk sharing deal issuance seems small in comparison, we must remember that the GSEs are only selling subordinate tranches, while the jumbo issuance displayed includes senior securities. If 8% of the jumbo deals issued were subordinate tranches, they would equate to ~\$1.1 billion of securities. So while overall issuance remains very low (as seen in Figure 2), the GSE transactions represent a sizable share of residential mortgage credit risk sold in 2013.

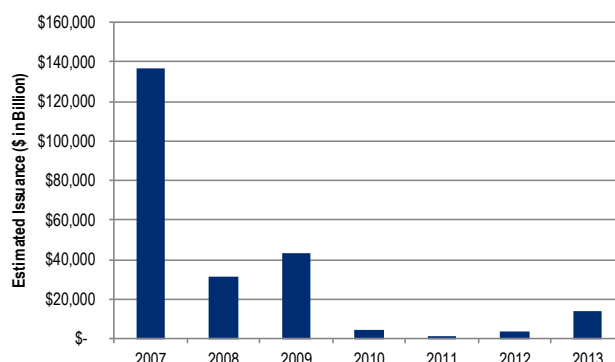
¹ <http://www.fhfa.gov/webfiles/25023/2013EnterpriseScorecard3413.pdf>

Figure 1. RMBS Issuance by Select Sectors



Source: CoreLogic, Intex, Citi Research Estimates

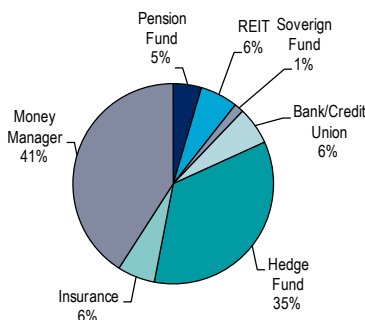
Figure 2. Prime Jumbo RMBS Issuance



Source: CoreLogic, Citi Research

Demand: Potential Buyers

Figure 3. STACR 2014-DN1 Investor Breakdown



Source: Citi Research, Freddie Mac estimates

Given the value profile (discussed later), money managers are probably the natural buyer of the M1 class. There is a potential for insurance companies as well, providing they can establish reasonable expected loss assumptions for NAIC rating purposes. According to the most recent NAIC data set that we have, these securities have yet to make it into the annual RMBS NAIC price breakpoint files. Banks are unlikely to be buyers given the thinness of the tranches, in our view, as the SSFA regulatory capital formula would be fairly punitive. Even with 0% delinquencies in many cases, even the M1 classes reach the highest risk weightings with the SSFA calculation. To the extent bank investors can get more comfortable with Volcker Rule Investment Act exemptions associated with re-REMICs, they may prove to be a more capital friendly investment. In the most recent Freddie Mac STACR transaction, Freddie estimates that money managers and hedge funds were the largest participating investor groups (Figure 3).

Comparison to Other Fixed Income Sectors

Figure 4 compares the STACR bonds to other comparable fixed income sectors. Looking at the M1, good comparables are higher-quality investment grade corporates and front sequential, shorter average life, re-REMICs. For the M2 tranche, we felt it best to compare to other sectors that bear some risk of loss and have longer average lives like high yield and prime hybrid ARMs. The yields we have used for comparison are based on spot rates. If the forward curve is realized, there is potential for higher yields, as seen in Figure 20.

Given the thinness of the tranches, there is more risk of outsized losses in tail scenarios. As a result, we would expect the STACR bonds to trade at some spread to other sectors such as corporates. Currently, it also appears that front pay, investment grade re-REMICs appear to have a more attractive loss-adjusted return.

Figure 4. Cross Sector Comparison

Tranche M1				Tranche M2			
Sector	Average Life (Yrs)	DM/Spread (bps)	Yield (%)	Sector	Average Life (Yrs)	DM/Spread (bps)	Yield (%)
IG Corporates 1-5 Yr	2.9	59	1.39	High Yield	6.6	311	5.23
BBB Corporates 1-5 Yr	3.0	87	1.70	High Yield, BB	7.1	212	4.38
Prime Front Pay Re-REMIC	2.0 - 4.0	120 - 150	1.60 - 1.80	High Yield, B	6.0	320	5.15
GSE M1 Classes	2.7 - 3.1	140 - 150	1.40 - 1.60	Prime Hybrid ARM PT*	6.0 - 7.0	215-225	4.00 - 5.00
				GSE M2 Classes	7.0 - 9.0	410 - 430	4.30 - 4.50

Source: Citi Research, the Yield Book as of 02/21/2014

* The yield shown for Prime Hybrid ARM PT is a loss adjusted yield

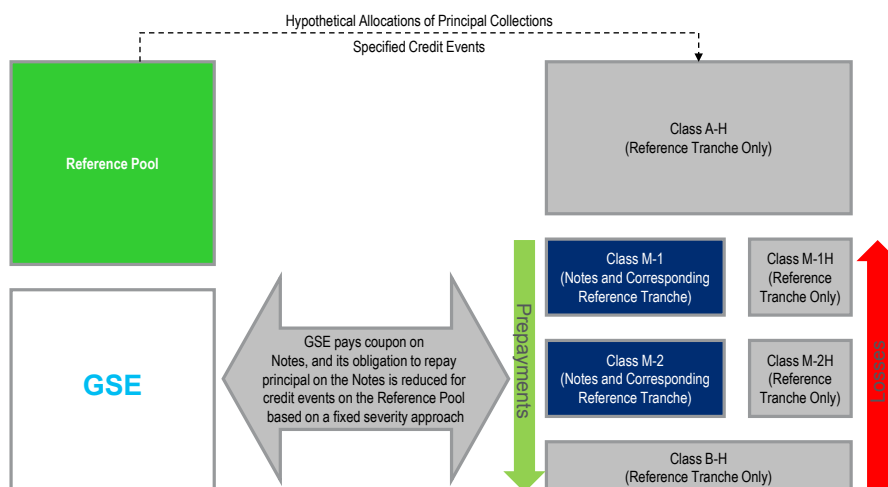
Bonds sold are structured as agency debt, cash flows are driven by a reference pool of mortgages. Bonds sold are uncapped LIBOR floaters.

Deal Construction: How They Are Structured

Generally speaking, cash flows to the bonds are based on the performance of reference loans, and the bonds are structured as general unsecured obligations of the GSE. In other words, bond payments will come from Fannie/Freddie and not passed through directly from cash flows generated by the mortgages, as we have traditionally seen in RMBS transactions. This is important, as this makes these securities agency debt, so their trading activity is all observable via FINRA's TRACE system. We will discuss trading activity later. This choice of deal structure is also important, as it retains the TBA eligibility of the underlying loans, giving up the liquidity of the TBA market would have been a significant negative for any sale of credit risk.

The dark blue sections of Figure 5 show the securities that are being sold. Initially there were two classes (M-1 and M-2), and subsequent transactions started to include exchangeable classes (starting with STACR 2013-DN2). The most recent Freddie Mac transaction (STACR 2014-DN1) includes three mezzanine tranches. The GSEs are required to buy back the notes at par in 10 years and have the option to call the transaction at 10% of the original balance. All bonds sold were structured as uncapped 1-month LIBOR-indexed floating rate bonds.

Figure 5. Generic Risk Sharing Deal Diagram



Source: Citi Research, Fannie Mae, Freddie Mac

In addition to the mezzanine tranches that were sold, there is also a 30bp thick first loss piece and a 97% thick, 3% credit enhancement senior piece that were retained by each GSE. The most recent Freddie Mac transaction, with its three mezz classes, provides 4.5% credit enhancement to the retained senior class. The GSEs also retained a portion of each mezzanine tranche (called M-1H and M-2H in Figure 5). They reserved the right to sell more of the mezzanine classes at some point in the future, but will retain at least 5% of each tranche. This 5% vertical slice retention specification is likely to align themselves with potential risk retention requirements in the future.

Some structural trends become evident when you stack up the transactions against each other, as seen in Figure 6. First, Fannie Mae acquired NRSRO ratings from Fitch, Moodys, or Kroll. Second, Freddie adjusted the thickness and credit enhancement levels of the M1/M2 classes (in STACR 2013-DN2) as well as adding a third M3 tranche (in STACR 2014-DN1). Finally, you can also see the different triggers and loss severity scales between CAS and STACR transactions.

Figure 6. Pricing Information for Risk Sharing Transactions Compared

Pricing Date	STACR 2013-DN1 7/23/2013	CAS 2013-CO1 10/22/2013	STACR 2013-DN2 11/7/2013	CAS 2014-CO1 1/14/2014	STACR 2014-DN1 2/6/2014
M1					
Rating (Fitch/Mdy/KBRA)	n/a	BBB-/NR/NR	BBB-/Baa1/NR	BBB-/Baa2/NR	NR/A1/A
Orig Size (\$ in '000)	250,000	337,500	245,000	375,000	240,000
% Thickness (% by bal)	1.35	1.35	1.05	1.35	1.00
Orig C/E (%)	1.65	1.65	1.95	1.65	3.50
Pricing DM (bps)	340	200	145	160	100
Pricing CPR (%)	10	10	10	10	10
Pricing WAL (yrs)	2.2	2.2	1.6	2.2	1.8
M2					
Rating (Fitch/Mdy)	n/a	n/a	n/a	n/a	NR/Baa1/BBB
Orig Size (\$ in '000)	250,000	337,500	385,000	375,000	360,000
% Thickness (% by bal)	1.35	1.35	1.65	1.35	1.50
Orig C/E (%)	0.30	0.30	0.30	0.30	2.00
Pricing DM (bps)	715	525	425	440	220
Pricing CPR (%)	10	10	10	10	10
Pricing WAL (yrs)	8.2	8.2	7.4	8.2	4.8
M3					
Rating (Fitch/Mdy)	n/a	n/a	n/a	n/a	n/a
Orig Size (\$ in '000)	n/a	n/a	n/a	n/a	408,000
% Thickness (% by bal)	n/a	n/a	n/a	n/a	1.71
Orig C/E (%)	n/a	n/a	n/a	n/a	0.30
Pricing DM (bps)	n/a	n/a	n/a	n/a	450
Pricing CPR (%)	n/a	n/a	n/a	n/a	10
Pricing WAL (yrs)	n/a	n/a	n/a	n/a	9.4
Cum Default Trigger? (Y/N)	Yes	No	Yes	No	Yes
Min C/E Trigger	3%	3%	3%	3%	5%
Loss Severity Scale:					
Cum Default Bucket					
0-1%	15%	10%	15%	10%	15%
1-2%	25%	20%	25%	20%	25%
>2%	40%	40%	40%	40%	40%

Source: Citi Research, Fannie Mae, Freddie Mac

Figure 7. Cumulative Default Trigger Values (STACR 2013-DN1)

Payment Date	Cumulative Net Credit Event % ("Cum Defaults")
Aug 2013 - Jul 2014	0.25%
Aug 2014 - Jul 2015	0.50%
Aug 2015 - Jul 2016	0.75%
Aug 2016 - Jul 2017	1.00%
Aug 2017 - Jul 2018	1.25%
Aug 2018 - Jul 2019	1.50%
Aug 2019 - Jul 2020	1.75%
Aug 2020 - Jul 2021	2.00%
Aug 2021 - Jul 2022	2.25%
Aug 2022 and thereafter	2.50%

Source: Citi Research

Figure 8. Loss Severity Schedule (STACR 2013-DN1)

Cum Defaults	Loss Severity (%)
≤ 1%	15
1 - 2%	25
> 2%	40

Source: Citi Research

Payment and Loss Allocation

Losses are allocated sequentially from the bottom of the capital structure, starting with the B-H class and moving up to the M-2 and then M-1. Unscheduled principal payments are paid pro rata between the senior and subordinate classes until any triggers are tripped, in which case all unscheduled principal goes to the senior A-H class. Scheduled principal is always paid pro-rata between the senior and subordinate classes. All principal payments to the subordinate classes are paid sequentially to the M-1, M-2, and then B-H classes. The sequential payment structure of the subordinate classes has been retained in all risk sharing deals.

Deal Triggers

There are two triggers in the deal that could cut off unscheduled principal to the sub classes:

1. The senior bond credit enhancement falls below 3%. [NOTE: The most recent STACR transaction minimum credit enhancement requirement is 5%.]
2. The cumulative defaults reach or exceed the trigger value. This trigger is on a sliding scale as described in Figure 7. [NOTE: This trigger was only used in the STACR risk sharing transactions.]

Treatment of Defaults, Losses, and Prepayments

Loans generally leave the pool via a credit event in several ways: reaching 6 months delinquent, short sale, deed-in-lieu of foreclosure, REO acquisition, and sale of a note to a third party during the foreclosure process. In all cases of a credit event, a fixed loss severity is used once loans are removed from the pool. The schedule of loss severity for increasing cumulative defaults is shown in Figure 8 [NOTE: Fannie Mae CAS transactions follow a 10/20/40 loss severity scale]. The GSEs reserve the right to reverse removals and write up bond balances if a loan is repurchased, as in the case of a rep&warrant violation putback.

In addition to voluntary prepayments, loans will leave the pool as a full prepayment if they are flagged as having an underwriting defect (in STACR transactions). Fannie Mae is slightly different, in that they remove loans when the servicer agrees to repurchase a loan or provides full indemnification. Loan modifications, if they don't meet any of the requirements to be removed, stay in the pool. However, any reduced principal through a modification will be considered a prepayment.

Fundamentals: Collateral Characteristics and Analysis

Generally speaking, the reference pools consist of amortizing 30-year fixed rate loans originated after mid-2012. Loans generally satisfy the following criteria:

1. 60-80% original LTV (Loan to Value Ratio)
2. No HARP loans (Home Affordable Refinance Program over 80% LTV refinances)
3. No MI (Mortgage Insurance)
4. No FHA, VA, or Rural Housing program loans
5. Never been 30 days or more delinquent before a pre-determined period
6. No known or unconfirmed underwriting defects

In line with the characteristics of recent GSE originations, the loans are high credit quality. Figure 9 provides some aggregate pricing statistics as well as some current characteristics according to recent remittance reports.

Collateral is High Quality, but Some Evidence of Risk Layering

Collateral characteristics are remarkably similar across transactions, and have only marginally more credit risk in subsequent transactions. Figure 10, Figure 11, and Figure 12 show the distributions of credit score (FICO), combined loan to value (CLTV), and debt to income ratio (DTI), respectively for the first three transactions.

Figure 13 and Figure 14 show the overall CLTV and DTI distribution compared to the distribution of 600-680 FICO loans for CAS 2014-C01. CLTV does not exhibit risk layering, but lower credit borrowers tend to have a higher DTI. This is consistent with other transactions.

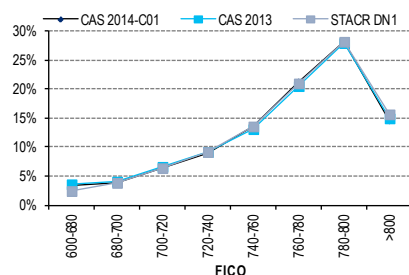
Figure 9. Origination and Current Loan Characteristics for Risk Sharing Deals

Orig Characteristics	STACR 2013-DN1	CAS 2013-C01	STACR 2013-DN2	CAS 2014-C01	STACR 2014-DN1
Loan Count	96,389	112,871	145,598	122,291	139,513
Loan Size	234,304	237,053	242,636	239,664	236,032
WAC	3.84	3.85	3.60	3.64	3.72
WAM	350	349	353	348	353
WALA	10	11	7	12	7
LTV	75	75	74	75	75
CLTV	76	76	75	76	76
DTI	32	32	32	32	33
FICO	766	765	764	765	761

Current Characteristics	STACR 2013-DN1	CAS 2013-C01	STACR 2013-DN2	CAS 2014-C01
Occupancy (%Curr)				
Owner Occupied	89.3	86.6	88.6	87
Amort Type (%Curr)				
Fixed	100	100	100	100
Geo (%Curr)				
%CA	22.4	27.2	24.5	30
%Judicial	39.8	35.6	37.7	22.8
Loan Purpose (%Curr)				
Refinance	42.8	58.9	53.2	68.6
Original Amort Term (%Curr)				
%30 Year	100	99.6	100	99.6
DQ Pipeline (%Curr)				
Current	99.8	99.9	99.9	100
D30	0.1	0.1	0.1	0

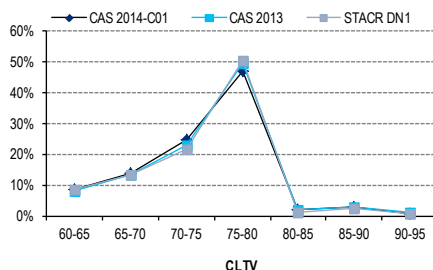
Source: Citi Research, Fannie Mae, Freddie Mac as of Jan 2014

Figure 10. FICO Distribution by Balance



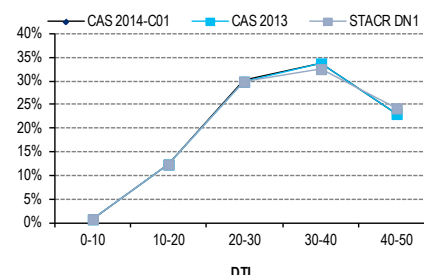
Source: Fannie Mae, Freddie Mac, Citi Research

Figure 11. Orig. CLTV Distribution by Balance



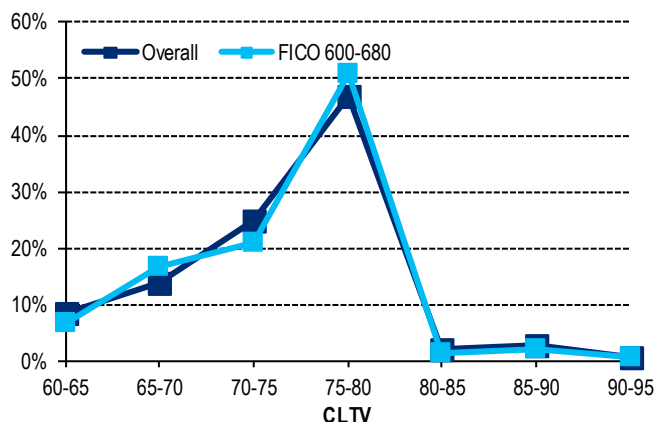
Source: Fannie Mae, Freddie Mac, Citi Research

Figure 12. DTI Distribution by Balance



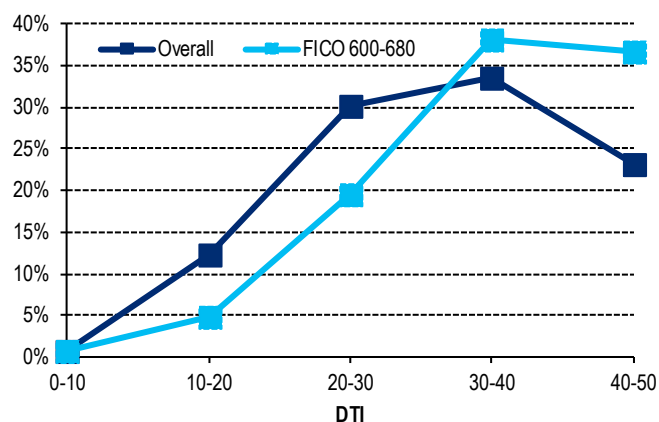
Source: Fannie Mae, Freddie Mac, Citi Research

Figure 13. CLTV / FICO Don't Show Risk Layering (CAS 2014-C01)



Source: Fannie Mae, Citi Research

Figure 14. Lower FICO Borrowers Exhibit Higher DTIs (CAS 2014-C01)



Source: Fannie Mae, Citi Research

Estimating Cumulative Defaults and Losses Using Historical Data

Figure 15 shows the aggregate characteristics of Freddie Mac originations by origination year. 2007 vintage loans had the highest cumulative defaults, reaching more than 12% 5 years after the origination date. In contrast, only 2.5% of 2003 vintage loans had defaulted 10 years after origination. There are 3 major factors that explain these differences: underwriting, home price appreciation, and prepayments.

There was a significant tightening in the underwriting standards after 2008. The average FICO at origination increased from 723 in 2007 to 762 in 2009 while the average CLTV dropped from 77 to 69 and the DTI declined from 37% to 33%.

Cashout refinances led to lower defaults on pre-2004 originations.

The pre-2004 originations had higher prepaids leading to their factoring down before the housing downturn. Hence the cumulative defaults on the pre-2004 originations were much lower than the 2005-08 originations. The 3 year cumulative HPA in 2002 and 2003 was in the mid to high 40% range, which created elevated cash out activity and higher prepayments. Without this refinancing, the defaults could have been much higher on pre-2004 originations.

Figure 15. Aggregate Characteristics of Freddie Loan Level Data

Orig Year	Balance (\$B)	CLTV	%Owner	DTI	Purchase	FTHB of Purchase	Cash out	Refi	FICO	Orig WAC	%CA	%Prepaid	% Defaults	3 year HPA
1999	129.6	77.5	94.3%	32.0	56.8%	22.5%	16.6%	26.6%	707.6	7.25	17.8%	87.4%	2.0%	30.9%
2000	98.4	79.0	92.5%	34.6	77.5%	22.9%	9.9%	12.6%	703.9	8.18	12.3%	83.6%	2.0%	30.8%
2001	245.0	76.5	93.8%	33.0	41.4%	21.3%	24.5%	34.1%	712.6	7.03	17.5%	85.9%	2.0%	37.3%
2002	263.0	74.9	93.5%	33.3	36.0%	21.0%	27.0%	37.0%	716.2	6.60	18.5%	70.7%	2.0%	46.4%
2003	322.1	73.4	94.2%	32.1	27.5%	19.3%	25.3%	47.2%	724.6	5.76	16.4%	62.4%	2.5%	42.2%
2004	188.0	75.5	92.4%	34.8	46.4%	18.4%	25.6%	28.0%	717.7	5.84	13.9%	59.2%	4.3%	19.3%
2005	236.8	74.1	92.8%	35.9	43.4%	20.3%	36.8%	19.8%	723.1	5.80	12.8%	52.6%	7.6%	-11.5%
2006	201.9	75.6	91.5%	37.1	50.7%	21.7%	34.6%	14.8%	722.5	6.41	9.9%	56.2%	10.6%	-25.4%
2007	201.7	77.1	89.8%	37.1	47.1%	26.0%	34.6%	18.3%	722.6	6.35	10.4%	49.2%	12.2%	-22.6%
2008	210.7	74.2	88.4%	37.1	41.9%	26.1%	31.0%	27.1%	737.6	6.11	15.7%	51.6%	7.7%	-16.0%
2009	344.6	69.0	92.9%	32.8	24.1%	31.2%	26.8%	49.1%	762.1	5.02	17.6%	38.7%	0.7%	-1.0%
2010	178.8	71.2	90.4%	33.4	37.4%	32.1%	24.7%	37.8%	761.8	4.84	21.4%	22.6%	0.2%	-5.5%
2011	134.0	72.4	89.7%	33.3	42.6%	29.4%	20.2%	37.2%	762.6	4.59	24.0%	7.7%	0.0%	0.0%
2012	14.5	71.3	91.8%	32.5	29.2%	28.4%	19.5%	51.2%	766.4	4.14	25.8%	0.0%	0.0%	0.0%

Source: Citi Research, CoreLogic

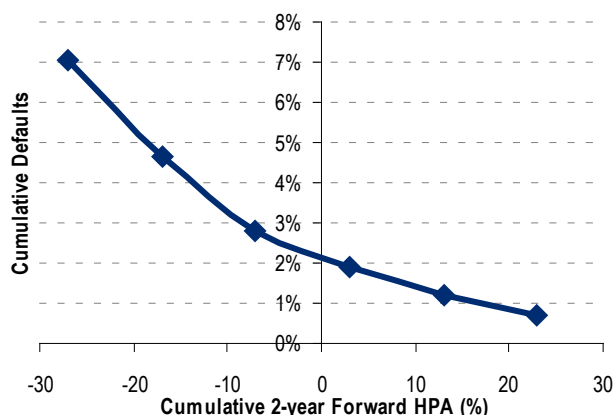
Note: HPA shown is based on National level CoreLogic HPI

We used the historical cumulative defaults on loans originated between 1999 and 2007 with similar characteristics as loans in the STACR 2013-DN1 deal to estimate the cumulative defaults under different HPA scenarios (Figure 16). Each loan in the STACR deal was mapped (based on FICO, CLTV, and DTI) to similar loans in the Freddie dataset and a default value was assigned based on how these loans performed under different HPA scenarios. Figure 17 shows the corresponding cumulative losses assuming the loss severity schedule in Figure 8.

The M1 tranches are unlikely to take losses.

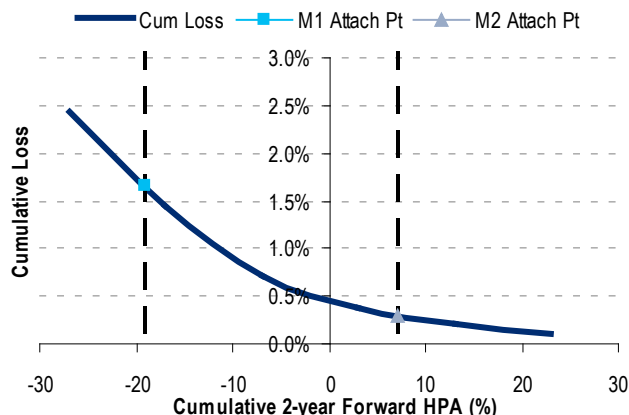
Based on this projection, tranche M2 will realize losses if the cumulative 2 year forward HPA is less than 7%. Tranche M1 will take losses if the cumulative 2 year forward HPA is about -19%. This is an arguably crude estimation that doesn't take into account other factors such as unemployment, but highlights our view that there is low credit risk associated with the M1 class and only marginal credit risk associated with the M2 class.

Figure 16. Cumulative Defaults In Different HPA Scenarios



Source: Citi Research

Figure 17. Cumulative Losses In Different HPA Scenarios



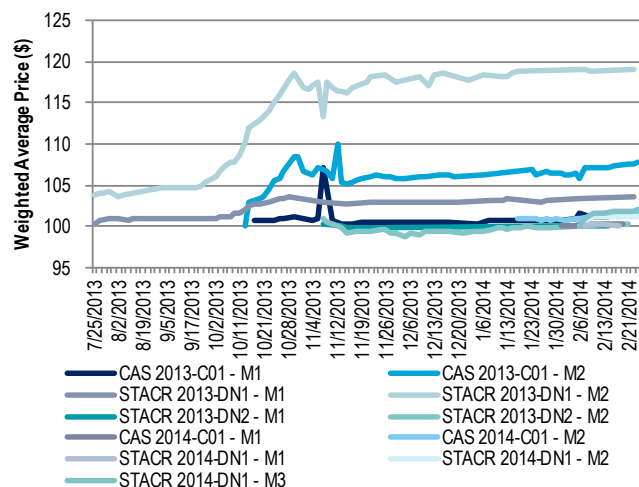
Source: Citi Research

Technicals: Secondary Trading Activity

Secondary pricing has been stable.

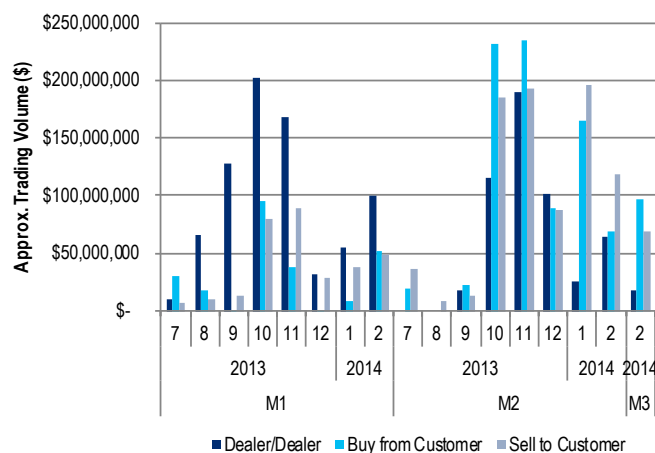
Since these securities are all considered agency debt, they are all reported on the TRACE system through FINRA. Figure 18 and Figure 19 highlight some secondary trading activity (price and volume) estimates from their website. You can see in the early price action that the market was still figuring out exactly what spread these bonds should trade to, but this has since subsided. Price stability shows that this market has matured quickly (Figure 18). Dealer-to-dealer trading volume dominated flows initially, but this has also fallen off and been taken over by more investor buying and selling, again a sign of a quickly maturing market (Figure 19).

Figure 18. Price Action on Risk Sharing Transactions



Source: Citi Research, FINRA

Figure 19. Trading Volume of Risk Sharing Transactions



Source: Citi Research, FINRA

Valuation Results Compared

Figure 20 highlights yield table results at current bid side prices using the Yield Book jumbo agency model. A couple observations are notable here:

The first transaction M1 class trades wider due to not having a rating.

The STACR 2013-DN1 M1 trades ~75 bps wider than other comparable M1 classes. This is primarily due to the fact that this class has no rating, and this spread differential is expected in our view. As a comparable, non-agency RMBS legacy front pay re-REMICs with an investment grade rating tend to trade anywhere between 50-100bps tighter than a non-rated comparable security.

Looking at the cumulative loss and default figures across the bottom of the table, you can see that cumulative loss estimates only break above 30bps, highlighting that current expectations for losses on all classes are relatively low.

Figure 20. Yield Table Results

	STACR 2013-DN1		CAS 2013-CO1		STACR 2013-DN2		CAS 2014-CO1		STACR 2014-DN1		
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2	M3
Bid Px	103-16	118-08	101-16	107-08	100-00	100-08	101-04	101-16	100-08	101-08	101-24
To Spot Rates:											
Yield	2.40	4.87	1.66	4.49	1.61	4.41	1.41	4.40	1.07	2.19	4.30
DM	224	467	150	429	145	422	125	421	92	202	411
To Forward Rates:											
Yield	3.46	6.99	2.71	6.76	2.50	6.77	2.57	6.81	1.91	4.35	6.82
DM	222	451	149	422	145	421	125	419	92	201	411
Duration	2.9	6.7	3.0	7.2	2.7	7.4	3.2	7.6	3.0	6.9	7.9
WAL	3.1	9.0	3.1	9.2	2.8	9.0	3.3	9.5	3.0	7.5	10.0
OAS	247	370	156	361	129	235	124	328	88	183	90
Eff Duration	-0.37	3.02	-0.05	2.76	0.37	5.36	0.12	3.60	0.31	0.89	9.15
1 yr VPR	5.5	5.5	5.7	5.3	4.5	4.5	5.2	5.2	4.1	4.1	4.1
Long Term VPR	7.4	7.4	7.4	7.3	6.8	6.8	7.0	7.0	6.8	6.8	6.8
1yr CDR	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Long Term CDR	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
10yr Cum Def (%Curr)	1.19	1.19	1.2	1.21	1.46	1.46	1.26	1.26	1.75	1.75	1.75
10 yr Cum Loss (% Curr)	0.14	0.18	0.09	0.13	0.15	0.25	0.1	0.15	0.16	0.33	0.33

Source: Citi Research, the Yield Book as of 02/21/2014

Other Applications of Pricing Data—Estimating Market-Implied G-Fees

We can use the pricing data to determine an implied guarantee fee.

The pricing data associated with these GSE risk sharing transactions also has other applications. One of these is to estimate what the market is telling us the GSE guarantee fees (g-fees) should be. In order to perform this exercise, we assume that the GSEs are going to hold capital against their loans in a similar fashion as banks. For the purposes of this analysis, we assume that the GSEs will hold approximately 5% of capital for these loans. Since they only sell a portion of the credit risk, we assume they will require a 5% return on capital for the remaining portion. The estimated implied fees are in Figure 21, also note that we include the reinsurance risk sharing transaction completed by Freddie Mac with Arch Reinsurance, Ltd. in November 2013 (see our discussion of this transaction in our [Mortgage Credit Weekly dated 14 November 2013](#)). We explain the g-fee estimation methodology more below, which is a refresh of a discussion in our [2014 Outlook](#).

The middle section of Figure 21 estimates the components of the guarantee fee. First, we have the payments made by the GSEs to the M1/M2/M3 classes. For example, the average coupon paid to the STACR 2013-DN1 deal is 527.5bps on 2.7% of the deal (rows a and b). This equates to 14.2bps of guarantee fee (row d, $527.5\text{bps} \times 2.7\%$).

We also assume the 30bps first loss piece is effectively an up-front payment and we use a multiple of 4 to turn it into a running fee (row e). For the other retained portion up to 5% credit enhancement, we assume the GSEs would require a 5% return on capital. This equates to 11.5bps in the case of STACR 2013-DN1 (row f, $5\% \times \text{row c}$).

We account for credit risk insurance payments, required return on capital as if the GSEs were banks, as well as operational costs.

Finally, if this type of risk transfer structure is going to be the ultimate method for the GSEs to operate on a standalone basis, one can't simply take insurance payments for the credit risk alone into account. The GSEs need to maintain the basic infrastructure of these transactions. Based on our review of recent GSE income statements, we estimate this operational cost around 10bps (row f), which is several times larger than a traditional trustee fee in non-agency RMBS transactions.

We estimate 48.5bps average implied g-fee based on risk sharing pricing data, compared with 57.4 bps on new origination.

After the Congressional mandated payroll tax cut surcharge of 10bps is added, we arrive to an average implied guarantee fee of ~48.5bps. This roughly matches current fees. According to Fannie Mae's recent 10-K filing, they are charging 57.4 bps on new origination. And it appears this current fee is covering their losses.

Figure 21. Estimating Market-Implied Guarantee Fees Using Risk Sharing Deal Pricing

Deal Name	STACR 2013-DN1	CAS 2013-C01	STACR 2013-DN2 Arch MI Transaction	CAS 2014-C01	STACR 2014-DN1
Pricing Date	7/26/2013	10/24/2013	11/12/2013	1/27/2014	2/6/2014
M1 Coupon (bps)	340	200	145	235	160
M2 Coupon (bps)	715	525	425	535	440
M3 Coupon (bps)					450
a) Premium Paid by GSE (average M1/M2, bps, wtd avg for STACR 2014-DN1)	527.5	362.5	285.0	385.0	300.0
b) % of Deal	2.7%	2.7%	2.7%	2.7%	2.7%
c) % of Deal up to 5% C/E (Bank Capital Requirement)	2.3%	2.3%	2.3%	2.3%	2.3%
Credit Risk:					
d) Insurance Premium paid on M1/M2 Credit Risk [a*b]	14.2	9.8	7.7	10.4	8.1
e) Up-Front Charge for 30bps First Loss Piece, Amortized (0.30% / 4)	7.5	7.5	7.5	7.5	7.5
5% Return on Capital on Unsold Portion:					
f) Return on Capital [5%*c]	11.5	11.5	11.5	11.5	11.5
g) Operational Costs (est. from FNMA 10-Q):	10.0	10.0	10.0	10.0	10.0
Estimated G-Fees:					
h) Total Estimated Guarantee Fee--Investor Perspective (bps) [d+e+f+g]	43.2	38.8	36.7	39.4	37.1
Total Estimated Guarantee Fee--GSE Perspective (bps) [h+10bps]	53.2	48.8	46.7	49.4	47.1

Source: Citi Research, Fannie Mae, Freddie Mac

Conclusion

Fannie Mae and Freddie Mac succeeded in their 2013 goal of transferring multiple forms of credit risk to the private market. They were all well-received by the marketplace, as seen in the rally in bond prices and tightening pricing spreads over 2013. Trading levels are stable, driven by investor activity. Now that the proof of concept has been established, we expect to see much more programmatic issuance over the course of 2014.

Appendix A-1

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